

# ROBOTIC ARM

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Constructing a robotic arm is a good start towards entering the amazing world of robotics. It involves the use of mechanics, physics, programming of microcontrollers and a lot more.

There are multiple approaches to design a robotic arm. The designer can choose between cost and precision performance. The mechanical construction of a robot is an aspect highly influenced by the way one wishes to control it.

Here we construct a robotic arm that has five degrees of freedom (DOF) and can move a load of up to 100 grams up to a distance of 35 cm. Various options are available ranging from powerful DC gear motors to more accurate servo motors.

Servos are normally used for moving the mechanical limbs of the robotic arm. This project uses seven servo motors for the movement of the robotic arm. Microcontroller PIC18F4431 from Microchip is used to control the servos. A keypad is connected to the microcontroller to manually control the motion of different parts of the arm.

## Basic servo operation and principle

A servo motor internally has a closed-loop feedback system that enables precise control of the position of the shaft of the motor. It also has a gear mechanism to generate torque. Servos usually have limited rotation (0-180 degrees). Six servos used in this project have a stall torque rating of 6 kg-cm at 6V and consume approximately 750 mA each. This means that the servo can lift a 6kg load to a height of 1 cm from the centre of its shaft when it's supplied 6V at 750 mA. The servo used for the gripper has a lower rating of 1.5 kg-cm at 6V.

To control the position of the servo motor, a pulse-width modulation (PWM) signal has to be supplied to it. The signal requirement may

vary for different servos. Varying the pulse-width from 0.8 millisecond to 2.2 milliseconds changes the servo shaft's position from 0 to 180 degrees. Fig. 1 shows PWM timing signals of the servo motor.

## Mechanical construction

The base of the arm is made up of an acrylic sheet cut according to the required shape and dimension. Fig. 2 shows the side view of the robotic arm

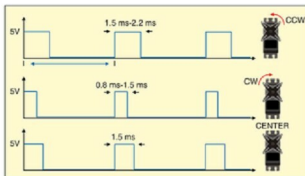


Fig. 1: Servo motor's PWM timing signals

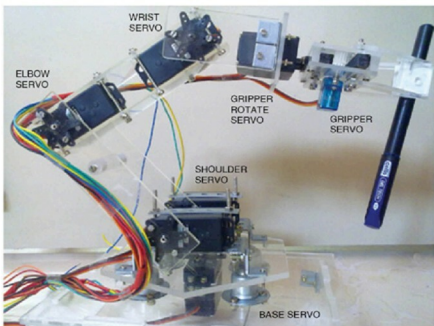


Fig. 2: Side view of the robotic arm holding an object

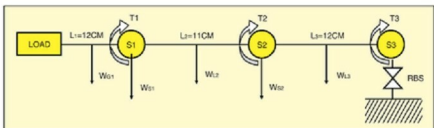


Fig. 3: Stick diagram representing the robotic arm